ADVANCE2000

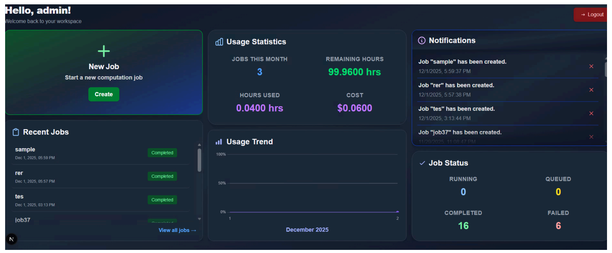
CLAIRVOYIX

AI GPU Power. Zero Complexity.

Anirudh Sanjay Mhaske | Marcus Hartman | Vidit Rajesh Prabhu

CLAIRVOYIX AT A GLANCE

Clairvoyix is Advance2000's next-generation private AI & GPU compute platform that makes high-performance computing as simple as using an app. It lets architects, engineers, analysts, and creators run GPU-accelerated AI workloads without any technical setup. Clairvoyix handles everything behind the scenes GPU allocation, containerization, model deployment, and performance tuning and gives users a clean, intuitive interface.



WHY CLAIRVOYIX?

- Data Integrity and Security
- Transparent Cost Control
- Zero-Setup Scalability
- Simple, unified user experience
- Scalable foundation

Core Narrative

Empowering organizations to harness AI compute privately and effortlessly

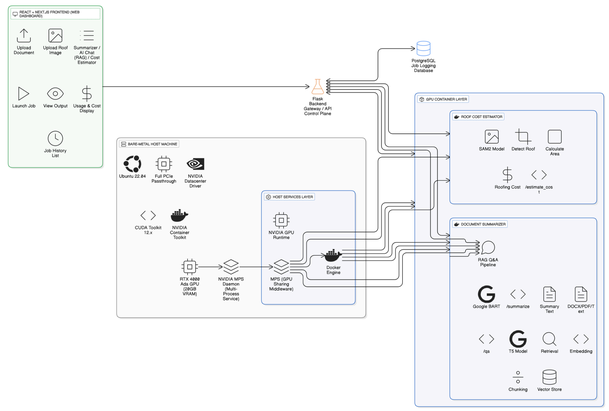
" Clairvoyix makes 'private AI as a service' real, secure, automated and human centered "

Why Clairvoyix ?

- No third party data sharing
- GPU accelerated computation
- Full control over data, logs and model inference lifecycle
- Scalable foundation

ARCHITECTURE

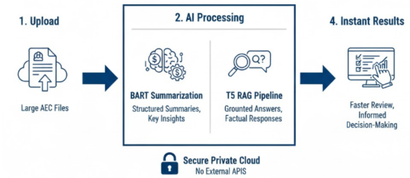
- Clairvoyix runs in a fully private GPU environment using an RTX 4000 Ada with PCIe passthrough, CUDA, and NVIDIA MPS for fast and concurrent AI workloads.
- Hosts all AI services inside Docker microservices, including BART summarization, T5 RAG-based Q&A, and SAM2 roof cost estimation.
- Uses a centralized Flask backend for job orchestration, request routing, GPU execution, and result handling.
- Provides a React and Next.js dashboard for file uploads, task launches, output viewing, usage tracking, and job history.
- Stores complete job activity in a PostgreSQL database, including inputs, outputs, GPU usage, costs, and lifecycle events.

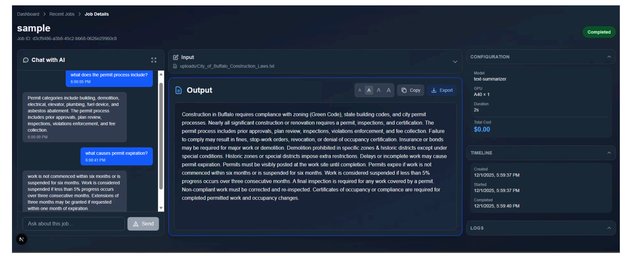


ENTERPRISE AI APPLICATIONS

AI Document Insights & Q&A

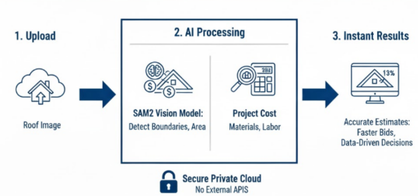
- Clairvoyix Instantly summarize large AEC documents and get grounded answers via our RAG pipeline, enabling faster review and smarter project decision-making.

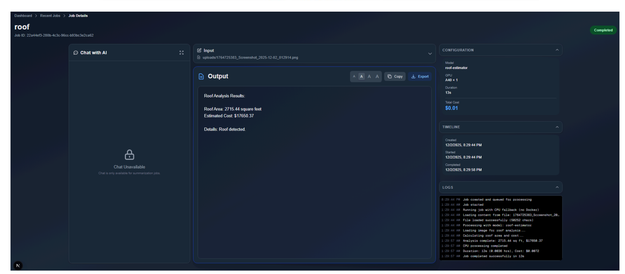




Roof Analysis & Cost Estimation

- Uses a SAM2-based vision model to analyze roof images, detect boundaries, compute surface area, and estimate project cost.





USER JOURNEY

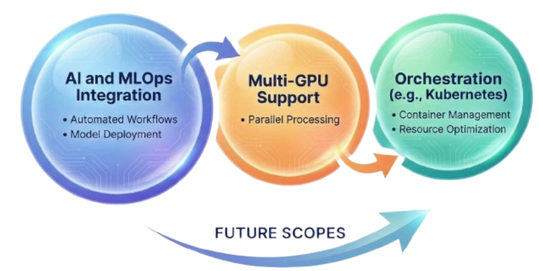
- **You Upload :** Send your document or roof image via the easy web dashboard.
- **We Compute:** Our private GPU system instantly processes your request.
- **You Get Results :** View summaries, answers, or cost estimates right away.
- **We Track :** See all job history, usage, and costs in one place.

PERFORMANCE OVERVIEW

- Private & Real-Time AI: Secure, real-time AI processing (analysis, estimation) delivered 100% within our private GPU cloud, eliminating external service reliance.
- Accuracy & Transparency: Provides accurate, grounded results (summaries, Q&A, cost predictions) with full job tracking and usage transparency on an integrated dashboard.

FUTURE ENHANCEMENTS

Moving forward, Clairvoyix will grow with additional AI services, integrated AI and MLOps pipelines, multi GPU support, and orchestration tools such as Kubernetes. These enhancements will allow the system to evolve into a fully scalable and comprehensive private AI compute platform.



University at Buffalo The State University of New York

Department of Computer Science and Engineering
School of Engineering and Applied Sciences
buffalo.edu